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Internship Report

Creating a validation pipeline using The Virtual Brain tool

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1 Introduction

Electroencephalography (EEG) is a fundamental neuroimaging technique characterized by excellent temporal resolution but limited spatial accuracy due to the volume conduction of the head tissues. To accurately estimate the underlying cortical sources (the *inverse problem*), a highly precise model of electrical propagation (the *forward problem*) is required. While numerical methods like the Boundary Element Method (BEM) are the gold standard for modeling this propagation, validating inverse algorithms requires a known *ground truth*, which can be synthetically generated using large-scale brain simulation platforms like *The Virtual Brain* (TVB).
This internship project focuses on developing, integrating, and validating an end-to-end computational pipeline for EEG source modeling. The core of the activity revolves around bridging TVB simulations with BEM solvers to evaluate source reconstruction algorithms. Specifically, the work involved:

- **Synthetic Data Generation:** Utilizing the TVB framework in Python to simulate biophysically realistic neural dynamics, ranging from static focal activations (Gaussian patches) to complex propagating epileptic seizures using the Epileptor model.
- **Forward Problem Modeling:** Computing the Leadfield matrix over a realistic three-layer head mesh (brain, skull, scalp) by implementing and comparing different Boundary Element Method (BEM) formulations, transitioning from Standard to an advanced Symmetric BEM.
- **Inverse Problem Validation:** Implementing a MATLAB-based validation framework to solve the inverse problem using algorithms such as sLORETA, Minimum Norm Estimates (MNE), and Weighted MNE (WMNE).
- **Performance Assessment:** Conducting extensive Monte Carlo simulations to evaluate the spatial and temporal accuracy of the reconstructions under varying Signal-to-Noise Ratios (SNR), using metrics like Localization Error (ED1) and Spatial Dispersion (ED2).

2 The forward problem in EEG

The scope of this section is to introduce the behavior of neurons inside the brain and to analyze different techniques for calculating the surface potentials (i.e., the EEG signals) generated by sources inside the brain.

2.1 Brief introduction to neuronal behavior

Understanding how EEG signals are generated requires a brief introduction to neurons [1]. All neurons have a basic structure: a central soma containing the nucleus, dendrites that receive signals, and an axon that sends signals to other cells.
However, the EEG cannot measure all electrical activity. Action potentials have a high voltage (70-110 mV), but they are too fast ($\approx$ 0.3 ms) to synchronize and be recorded on the scalp. Because of this, the EEG signal is mainly generated by postsynaptic potentials, which are slower and last longer (10-20 ms).
Importantly, the EEG does not record random activity from the whole brain. It records the synchronized activity of a specific group of cells: the pyramidal neurons.
Unlike other neurons that have random directions, pyramidal cells have long dendrites that are parallel to each other and perpendicular to the cortex surface. This parallel structure allows their electrical currents to sum up, creating the large macroscopic signals we see in the EEG.

2.2 Quasi-static conditions and general hypotheses

For the frequency range of the signals measured in the EEG, no charge can pile up in the conducting extracellular volume. Therefore, at any given moment, all fields and currents behave as if they were stationary; these are called quasi-static conditions. While the neural activity changes over time, these changes are slow compared to electromagnetic propagation effects [2].
Under these conditions, the current density $\mathbf{J}$ in regions beyond the site of an action potential is linearly related to the electric field intensity $\mathbf{E}$ [3, 4]. Furthermore, the capacitive component of tissue impedance is negligible at the frequencies of interest.
To a good approximation, a region containing no bioelectric sources can be assigned a uniform bulk conductivity σ, yielding Ohm's law:

$$\mathbf{J} = \sigma \mathbf{E} \tag{1}$$

where $\sigma(\mathbf{r}) \in \mathbb{R}^{3\times3}$ is the position-dependent conductivity tensor [1]. If the medium is isotropic, the tensor simplifies to a scalar value. In its matrix form, it is represented as:

$$\sigma = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_{22} & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} \end{bmatrix} \tag{2}$$

Because electromagnetic wave effects can be neglected under the quasi-static approximation, the time derivative of the magnetic field is negligible. According to Faraday's law of induction, this implies that the electric field is essentially irrotational ($\nabla \times \mathbf{E} = 0$). So, at each instant, the electric field can be obtained directly from the negative gradient of an electric scalar potential, V:

$$\mathbf{E} = -\nabla V \tag{3}$$

2.3 Poisson's equation

Formally, we can represent the active neuronal sources by a distribution of impressed current density, $\mathbf{J}^i$ [3, 4]. According to this formalism, Ohm's law is modified to include these active regions as follows:

$$\mathbf{J} = \sigma\mathbf{E} + \mathbf{J}^i \tag{4}$$

By combining (4) with (3) and the condition that $\nabla \cdot \mathbf{J} = 0$ (no charge accumulation), we obtain Poisson's differential equation in its general form:

$$\nabla \cdot (\sigma \nabla V) = \nabla \cdot \mathbf{J}^i \tag{5}$$

2.4 Boundary conditions

The head is not a single homogeneous mass, but consists of different compartments (e.g., brain, skull, scalp). Let the surface S_j separate regions of conductivity σ' and σ'', and let $d\mathbf{S}_j$ be a differential element of area on this surface directed from the primed region to the double-primed region. We have two essential boundary conditions that govern these interfaces [3].
First, the normal component of the current density must be continuous across each boundary:

$$\sigma' \nabla V' \cdot d\mathbf{S}_j = \sigma'' \nabla V'' \cdot d\mathbf{S}_j \tag{6}$$

Second, the electric potential cannot have discontinuities across an interface:

$$V'(S_j) = V''(S_j) \tag{7}$$

2.5 The current dipole and general algebraic formulation

Active neuronal groups can be modeled macroscopically as an equivalent current dipole [1]. As established, the potential generated by a source in an unbounded homogeneous medium is given by the solution to Poisson's equation:

$$V(\mathbf{r}) = \frac{1}{4\pi\sigma} \int_G \mathbf{J}^i \cdot \nabla \left(\frac{1}{\mathbf{R}} \right) d\nu \tag{8}$$

where G contains the source $\mathbf{J}^i$.
Given the physical behavior of neurons, we can consider our sources to be current dipoles. A current dipole with a moment $\mathbf{d}$ represents a concentration of the impressed current at a single mathematical point $\mathbf{r}_{\text{dip}}$, which can be expressed using the Dirac delta function as $\mathbf{J}^i(\mathbf{r}) = \mathbf{d}\delta(\mathbf{r} - \mathbf{r}_{\text{dip}})$. Substituting this into (8) collapses the integral to that specific point, yielding:

$$V(\mathbf{r}, \mathbf{r}_{\text{dip}}, \mathbf{d}) = \frac{1}{4\pi\sigma} \mathbf{d} \cdot \frac{\mathbf{r} - \mathbf{r}_{\text{dip}}}{\|\mathbf{r} - \mathbf{r}_{\text{dip}}\|^3} \tag{9}$$

By leveraging the linearity of the equation, the potential at an arbitrary scalp measurement point $\mathbf{r}$ due to this dipole can be decomposed along the Cartesian axes [1]:

$$V(\mathbf{r}, \mathbf{r}_{\text{dip}}, \mathbf{d}) = d_x V(\mathbf{r}, \mathbf{r}_{\text{dip}}, \mathbf{e}_x) + d_y V(\mathbf{r}, \mathbf{r}_{\text{dip}}, \mathbf{e}_y) + d_z V(\mathbf{r}, \mathbf{r}_{\text{dip}}, \mathbf{e}_z) \tag{10}$$

This can be further simplified by assuming the magnitude d to be equal in all directions.
The EEG forward problem consists of finding the scalp potential generated by these sources. For N electrodes and p dipoles across T discrete time samples, the forward problem is aggregated into a single matrix equation:

$$\mathbf{V} = \begin{bmatrix} V(\mathbf{r}_1,1) & \cdots & V(\mathbf{r}_1,T) \\ \vdots & \ddots & \vdots \\ V(\mathbf{r}_N,1) & \cdots & V(\mathbf{r}_N,T) \end{bmatrix} = \mathbf{G}\left(\left\{\mathbf{r}_j, \mathbf{r}_{\mathrm{dip}_i}, \mathbf{e}_{d_i}\right\}\right) \begin{bmatrix} d_{1,1} & \cdots & d_{1,T} \\ \vdots & \ddots & \vdots \\ d_{p,1} & \cdots & d_{p,T} \end{bmatrix} = \mathbf{GD} \tag{11}$$

Here, $\mathbf{V}$ is the $N \times T$ matrix of measured potentials, $\mathbf{D}$ contains the dipole magnitudes over time, and $\mathbf{G}$ is the lead-field matrix containing the forward transfer-coefficients mapping the sources to the sensors. More generally, a noise or perturbation matrix $\mathbf{n}$ is added to reflect realistic recording conditions [1]:

$$\mathbf{V} = \mathbf{GD} + \mathbf{n} \tag{12}$$

3 The Standard Boundary Element Method (BEM)

The Boundary Element Method (BEM) is a numerical technique for calculating the surface potentials generated by current sources located in a piecewise homogeneous volume conductor [1].
We begin by applying Green's theorem [3, 4]. By substituting $\psi = 1/\mathbf{R}$ (where $\mathbf{R}$ is the distance vector from the observation point to the integration element) and $\phi = V$, an implicit expression for the potential V is derived:

$$4\pi\sigma V\Big|_{\mathbf{R}=0} = \int_G \mathbf{J}^i \cdot \nabla\left(\frac{1}{\mathbf{R}}\right) d\nu - \sum_j \int_{S_j} V(\sigma' - \sigma'')\nabla\left(\frac{1}{\mathbf{R}}\right) \cdot d\mathbf{S}_j \tag{13}$$

As explicitly stated in the equation, the terms σ and V on the left side are evaluated at $\mathbf{R} = 0$, which is exactly the observation point where we want to measure the potential. The total potential in the head is the sum of the potential that the primary sources would produce if they were in an infinite homogeneous medium, plus the effect of "fictitious" secondary sources located on the boundary surfaces S_j where the conductivity changes.
Let $\mathbf{r}$ be the observation point and $\mathbf{r}'$ be the integration point on the surfaces. The distance vector is then $\mathbf{R} = \mathbf{r} - \mathbf{r}'$. Evaluating the left side at $\mathbf{R} = 0$ implies calculating the potential and conductivity at the observation point, $V(\mathbf{r})$ and $\sigma(\mathbf{r})$.
The first integral represents the potential $V_0(\mathbf{r})$ generated by the primary sources in an infinite homogeneous medium with conductivity σ_0. As the observation point $\mathbf{r}$ approaches a point on the boundary surface S_k from the inside, the surface integral exhibits a mathematical singularity. Resolving this limit yields an average conductivity term of $\frac{\sigma'_k + \sigma''_k}{2}$ [4].
By substituting these elements, we derive the standard BEM integral equation [1]:

$$V(\mathbf{r}) = \frac{2\sigma_0}{\sigma_k^- + \sigma_k^+} V_0(\mathbf{r}) - \frac{1}{2\pi} \sum_j \frac{\sigma_j^- - \sigma_j^+}{\sigma_k^- + \sigma_k^+} \int_{\mathbf{r}' \in S_j} V(\mathbf{r}') \frac{\mathbf{r} - \mathbf{r}'}{\|\mathbf{r} - \mathbf{r}'\|^3} \cdot d\mathbf{S}_j \tag{14}$$

3.1 Numerical Implementation and the Solid Angle

The BEM resolves the integral equation by restricting calculations to the interfaces (e.g., brain-skull, skull-scalp, scalp-air). Each interface S_j is tessellated into N_{S_j} small planar triangles. The integral over the entire surface S_j is thus transformed into a discrete summation of integrals over individual triangles:

$$V(\mathbf{r}) = \frac{2\sigma_0}{\sigma_r^- + \sigma_r^+} V_0(\mathbf{r}) - \frac{1}{2\pi} \sum_k^R \frac{\sigma_k^- - \sigma_k^+}{\sigma_r^- + \sigma_r^+} \sum_j^{N_{S_k}} \int_{\Delta_{S_{k,j}}} V(\mathbf{r}') \frac{\mathbf{r} - \mathbf{r}'}{\|\mathbf{r} - \mathbf{r}'\|^3} \cdot d\mathbf{S}_k \tag{15}$$

where the integral is performed over $\Delta_{S_{k,j}}$, the j-th triangle on the surface S_k, and R is the total number of interfaces.
In the Standard BEM approach, the potential is assumed to be constant over each specific triangle. This allows the potential V_i^k to be factored out of the integral. The remaining geometric integral corresponds to the solid angle subtended by the triangle at the observation point. An analytical formula for the solid angle Ω is given by [5]:

$$\tan\left(\frac{1}{2}\Omega\right) = \frac{\mathbf{R}_1 \cdot (\mathbf{R}_2 \times \mathbf{R}_3)}{R_1 R_2 R_3 + (\mathbf{R}_1 \cdot \mathbf{R}_2) R_3 + (\mathbf{R}_1 \cdot \mathbf{R}_3) R_2 + (\mathbf{R}_2 \cdot \mathbf{R}_3) R_1} \tag{16}$$

where $\mathbf{R}_1$, $\mathbf{R}_2$, and $\mathbf{R}_3$ are vectors pointing from the observation point to the three vertices of the triangle, and R_1, R_2, R_3 are their respective norms.
This approximation transforms the integral equation into a set of linear algebraic equations:

$$\mathbf{V} = \mathbf{B}\mathbf{V} + \mathbf{V}_0 \tag{17}$$

However, this system is rank-deficient and does not possess a unique solution. This ambiguity is removed through a mathematical technique known as deflation, replacing $\mathbf{B}$ with $\mathbf{C} = \mathbf{B} - \frac{1}{N}\mathbf{e}\mathbf{e}^T$, where $\mathbf{e}$ is a vector of ones. Defining $\mathbf{A} = \mathbf{I} - \mathbf{C}$ and defining $\mathbf{K}$ as the source contribution matrix:

$$\mathbf{K} = \frac{1}{2\pi\left(\sigma_r^- + \sigma_r^+\right)} \begin{bmatrix} \frac{\mathbf{R}_1}{\|\mathbf{R}_1\|^3} \\ \vdots \\ \frac{\mathbf{R}_N}{\|\mathbf{R}_N\|^3} \end{bmatrix} \begin{bmatrix} \mathbf{e}_{d_1} & \cdots & \mathbf{e}_{d_{N_s}} \end{bmatrix} \tag{18}$$

The final Leadfield matrix $\mathbf{G}$ is obtained by projecting the potentials to the sensor locations using a selection matrix $\mathbf{S}$:

$$\mathbf{G} = \mathbf{S}\mathbf{A}^{-1}\mathbf{K} \tag{19}$$

4 The Symmetric Boundary Element Method

While the Standard BEM is computationally straightforward, it suffers from severe numerical inaccuracies when dealing with realistic head geometries. Specifically, it is highly sensitive to the high conductivity contrast between the brain and the skull, and it produces significant errors when applied to thin layers, such as the human skull [6]. To overcome these limitations, an advanced formulation known as the Symmetric Boundary Element Method (Symmetric BEM) was adopted in this project.

4.1 Representation Theorem and Integral Operators

The Symmetric BEM is based on an application of the Representation Theorem [6]. To maintain a general mathematical formulation, let u be a harmonic scalar field (representing the electric potential) and let p be its associated normal flux across a boundary $\partial\Omega$, defined as:

$$p(\mathbf{r}) = \partial_{\mathbf{n}} V(\mathbf{r}) \tag{20}$$

where $\mathbf{n}$ is the outward-pointing unit normal vector at the surface point $\mathbf{r}$.
To express u and p on the boundaries, we define the fundamental solution (Green's function) for the Laplace equation in a 3D unbounded medium:

$$G(\mathbf{r}, \mathbf{r}') = \frac{1}{4\pi \|\mathbf{r} - \mathbf{r}'\|} \tag{21}$$

Based on this Green's function, we introduce four fundamental boundary integral operators mapping scalar functions on a surface $\partial\Omega$ to another surface (or the same surface) [6]:

- The **Single-layer operator** ($\mathcal{S}$):

$$(\mathcal{S}f)(\mathbf{r}) = \int_{\partial\Omega} G(\mathbf{r}, \mathbf{r}') f(\mathbf{r}') \mathrm{d}s(\mathbf{r}') \tag{22}$$

- The **Double-layer operator** ($\mathcal{D}$):

$$(\mathcal{D}f)(\mathbf{r}) = \int_{\partial\Omega} \partial_{\mathbf{n}'} G(\mathbf{r}, \mathbf{r}') f(\mathbf{r}') \mathrm{d}s(\mathbf{r}') \tag{23}$$

- The **Adjoint double-layer operator**:

$$(\mathcal{D}^* f)(\mathbf{r}) = \int_{\partial\Omega} \partial_{\mathbf{n}} G(\mathbf{r}, \mathbf{r}') f(\mathbf{r}') \mathrm{d}s(\mathbf{r}') \tag{24}$$

- The **Hypersingular operator**:

$$(\mathcal{N}f)(\mathbf{r}) = \int_{\partial\Omega} \partial^2_{\mathbf{n},\mathbf{n}'} G(\mathbf{r}, \mathbf{r}') f(\mathbf{r}') \mathrm{d}s(\mathbf{r}') \tag{25}$$

According to the Representation Theorem, any harmonic function u and its flux p inside or outside a domain Ω can be fully determined by their values on the boundary $\partial\Omega$. We obtain the following coupled system of integral equations:

$$\begin{aligned} -p &= +\mathcal{N}[u] && -\mathcal{D}^*[p] && \textit{for } \mathbf{r} \notin \partial\Omega \\ u &= -\mathcal{D}[u] && +\mathcal{S}[p] && \\ -p^{\pm} &= +\mathcal{N}[u] && + \left(\pm\frac{\mathcal{J}}{2} - \mathcal{D}^*\right)[p] && \textit{for } \mathbf{r} \in \partial\Omega \\ u^{\pm} &= \left(\mp\frac{\mathcal{J}}{2} - \mathcal{D}\right)[u] && +\mathcal{S}[p] && \end{aligned}$$

where $\mathcal{J}$ denotes the identity operator.

4.2 The Symmetric Formulation and Galerkin Discretization

Let us consider the standard three-layer head model with nested surfaces $\partial\Omega_1$ (brain-skull), $\partial\Omega_2$ (skull-scalp), and $\partial\Omega_3$ (scalp-air), separating regions with conductivities σ_1, σ_2, σ_3, and σ_4 (where the air conductivity is $\sigma_4 = 0$).
By evaluating the Representation Theorem limits approaching the boundaries from both the inside and the outside, and enforcing the continuity conditions for both the potential u and the normal flux p across each interface, we obtain a set of coupled boundary integral equations. Unlike the traditional BEM which leads to an asymmetric dense matrix, this coupled system can be arranged into a symmetric block structure. For the three-layer model, the system pairs the integral operators to compute the unknown potentials u_1, u_2, u_3 and fluxes p_1, p_2 on the respective surfaces (noting that $p_3 = 0$ since current cannot flow into the air).
To solve this continuous system numerically, the Symmetric BEM employs a Galerkin approach. The continuous variables are projected onto two distinct discrete functional spaces defined over the triangular mesh:

- **Pyramid functions:** The continuous potential u is approximated using piecewise linear basis functions associated with the vertices of the mesh.
- **Patch functions:** The discontinuous normal flux p is approximated using piecewise constant basis functions associated with the triangular faces of the mesh.

This specific Galerkin discretization leads to a global, symmetric block matrix equation $\mathbf{Zx} = \mathbf{b}$. For a standard three-layer head model with conductivities σ_1, σ_2, σ_3, and $\sigma_4 = 0$, and assuming that the normal flux vanishes at the outermost boundary ($p_3 = 0$), the full linear system expands to:

$$\begin{bmatrix} (\sigma_1+\sigma_2)\mathcal{N}_{11} & -2\mathcal{D}_{11}^* & -\sigma_2\mathcal{N}_{12} & \mathcal{D}_{12}^* & 0 \\ -2\mathcal{D}_{11} & (\sigma_1^{-1}+\sigma_2^{-1})\mathcal{S}_{11} & \mathcal{D}_{12} & -\sigma_2^{-1}\mathcal{S}_{12} & 0 \\ -\sigma_2\mathcal{N}_{21} & \mathcal{D}_{21}^* & (\sigma_2+\sigma_3)\mathcal{N}_{22} & -2\mathcal{D}_{22}^* & -\sigma_3\mathcal{N}_{23} \\ \mathcal{D}_{21} & -\sigma_2^{-1}\mathcal{S}_{21} & -2\mathcal{D}_{22} & (\sigma_2^{-1}+\sigma_3^{-1})\mathcal{S}_{22} & \mathcal{D}_{23} \\ 0 & 0 & -\sigma_3\mathcal{N}_{32} & \mathcal{D}_{32}^* & \sigma_3\mathcal{N}_{33} \end{bmatrix} \begin{bmatrix} u_1 \\ p_1 \\ u_2 \\ p_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ c_1 \\ b_2 \\ c_2 \\ b_3 \end{bmatrix} \tag{26}$$

where $\mathbf{Z}$ is the symmetric block matrix containing the numerical integrations of the $\mathcal{S}$, $\mathcal{D}$, $\mathcal{D}^*$, and $\mathcal{N}$ operators evaluated between the nested boundaries using the basis functions. The vectors $\mathbf{b}$ and $\mathbf{c}$ on the right-hand side represent the primary potential and flux generated by the cortical sources in an infinite homogeneous medium.

4.3 From the Symmetric System to the Leadfield Matrix G

To construct the final Leadfield matrix $\mathbf{G}$, the symmetric linear system must be solved for the entire set of N_s cortical dipoles. The individual right-hand side vectors $\mathbf{b}_n$ are aggregated into a global Right-Hand Side matrix $\mathbf{B} \in \mathbb{R}^{M \times N_s}$, where $M = N_{v_1} + N_{f_1} + N_{v_2} + N_{f_2} + N_{v_3}$ represents the total number of degrees of freedom across the layers.
The full electromagnetic state matrix $\mathbf{X}$ is computed by solving the deflated linear system:

$$\mathbf{X} = \mathbf{Z}_{\text{def}}^{-1}\mathbf{B} \tag{27}$$

Since EEG sensors only record the electric potential on the outermost boundary (the scalp), we extract the submatrix corresponding to the u_3 block, which contains the potentials evaluated at the N_{v_3} scalp vertices:

$$\mathbf{U}_3 = \mathbf{X}(1 + N_{v_1} + N_{f_1} + N_{v_2} : M, :) \tag{28}$$

Physically, ideal EEG measurements are inherently non-local and are determined up to an arbitrary reference potential. To reflect this recording property and ensure physical consistency across simulations, a common

average reference (CAR) is applied by subtracting the spatial mean of the potentials for each source configuration (column-wise):

$$\mathbf{G}_{\text{mesh}} = \mathbf{U}_3 - \frac{1}{N_{v_3}}\mathbf{e}\mathbf{e}^T\mathbf{U}_3 \tag{29}$$

where $\mathbf{e}$ is a column vector of ones of length N_{v_3}.

5 Large-Scale Brain Modeling with The Virtual Brain

5.1 What is TVB

The Virtual Brain (TVB) is an open-source neuroinformatics framework designed for the simulation of large-scale brain network dynamics with biologically realistic connectivity. Unlike microscopic, cellular-level simulators, TVB operates at the macroscopic level. The core concept relies on coupling a structural connectivity matrix—which defines the connection strengths and time delays for the signal transmission between all network nodes—with Neural Mass Models (NMMs) that describe the average population behavior at each node.
Together, the NMMs and the connectivity matrix define the Virtual Brain. TVB can simulate the time courses of various forms of neural activity, including Local Field Potentials (LFP) and firing rates, and directly project them to simulate empirical brain imaging data like EEG, MEG, and BOLD activations for fMRI.

5.2 The Data Processing Pipeline

To build a personalized Brain Network Model (BNM), empirical neuroimaging data must be processed through an automated pipeline to extract structural and functional connectomes, reducing high-dimensional imaging data into TVB-compatible formats [7]. The pipeline integrates several neuroinformatics tools, including FREESURFER, FSL, MRTrix, and Brainstorm. Usually, the required inputs are a T1-weighted structural MRI, BOLD fMRI data, and Diffusion-weighted MRI (dwMRI) with an adequate number of distinct gradient directions.
First, all data goes through standard preprocessing steps such as motion correction, intensity normalization, and skull stripping. Then, empirical functional data is processed to generate the targets that the TVB simulations will attempt to fit. Finally, if empirical EEG data is provided, it can be projected back to the cortical surface using a Boundary Element Method (BEM) forward model and an inverse solution like sLORETA. This empirical source-level activity provides a crucial benchmark for direct comparison with TVB's simulated cortical activity.

5.3 Co-simulations and the EBRAINS Ecosystem

While the standard TVB framework excels at simulating macroscopic whole-brain dynamics using Neural Mass Models, some research questions require exact cellular-level resolution while preserving the global network context. To address this, a co-simulation approach is often adopted, running TVB for the whole brain alongside a spiking neural network (SNN) simulator, such as NEST, for specific localized regions simultaneously. This computationally demanding setup is facilitated by the EBRAINS infrastructure, which provides HPC cloud environments and optimized pre-configured toolchains.
The microscopic data generated by a TVB-NEST co-simulation (e.g., discrete spike times and membrane potentials) is typically analyzed using the Electrophysiology Analysis Toolkit (Elephant). This allows the estimation of the Local Field Potential (LFP) directly from the spiking activity, effectively bridging the gap between micro and macro scales.

5.4 Fast TVB and High-Performance Computing (HPC)

Simulating a Brain Network Model involves numerically integrating a system of coupled stochastic non-linear differential equations. For a region-based simulation (e.g., 60-100 parcellated nodes), standard Python implementations are usually sufficient. However, for a surface-based simulation with tens of thousands of vertices, the computational complexity scales drastically due to the dense coupling matrix computations.
To solve these bottlenecks, the Fast TVB architecture translates the core numerical integration loops into low-level C/C++ code. This HPC optimization is fundamental for Parameter Space Exploration (PSE). The ultimate goal of personalized brain modeling is to find the specific biophysical parameters (like the global coupling scaling factor G and the conduction velocity v) that optimally fit the empirical data. PSE requires launching thousands of independent simulations across a computing cluster, making fast integration schemes mandatory.

6 Clinical Applications and Case Studies

6.1 Epilepsy: The Virtual Epileptic Patient

The main clinical application of TVB for epilepsy is the Virtual Epileptic Patient, which tries to predict patient-specific seizure propagation and find the surgical target [8]. Each network node is equipped with the Epileptor, a phenomenological model with fast, slow, and ultra-slow state variables across different time scales.

The spatial heterogeneity of the epileptic brain is modeled by mapping an excitability parameter x_0 across the network. Regions where x_0 exceeds a critical threshold act autonomously as the Epileptogenic Zone (EZ). Using a Bayesian inference framework, the simulated seizures are projected onto virtual sensors and fitted against the patient's empirical Stereotactic EEG (SEEG) recordings. This helps to estimate the exact spatial distribution of x_0, giving clinicians a probability map of the EZ [9].

6.2 Alzheimer's Disease (AD) Modeling

Modeling Alzheimer's Disease in TVB involves linking microscopic molecular pathways to global network degradation [10]. TVB tracks the progressive degradation focused initially on a Limbic SubNetwork. By using the spatial distribution of Amyloid-beta burden extracted from individual PET scans, TVB models its neurotoxic effect on the dopaminergic system.
Specifically, the disruption of GABA release is implemented via a transfer function that spatially modulates the inhibitory time constant τ_i. A higher regional Amyloid-beta load forces a slower filter action, causing local hyperexcitation. This propagates through the structural connectome, successfully reproducing empirical AD hallmarks like the global slowing of the virtual EEG rhythms from the healthy alpha band down to pathological theta and delta frequencies [11]. Virtual pharmacodynamics can also be tested, for instance simulating the effect of Memantine to functionally reverse the network alterations.

6.3 Stroke and Brain Tumor Modeling

For stroke recovery modeling, TVB evaluates the chronic cortical phase. Large strokes physically deform the brain, so a virtual transplant is performed: the lesioned area on the MRI is segmented and replaced with homologous healthy tissue. Simulations show that stroke patients exhibit a consistent reduction in conduction velocity and local inhibitory coupling, metrics that strongly correlate with long-term motor recovery scores [12].
In the context of presurgical planning for brain tumors, TVB models the non-linear network dynamics induced by gliomas and meningiomas [13]. Using Feedback Inhibition Control (FIC), the local feedback inhibition parameter is optimized to clamp the average firing rate. The optimized local parameters successfully differentiate between regions affected by the tumor and healthy tissue, significantly improving the prediction accuracy of the patient's empirical functional connectivity.

7 Adopted Models and Simulation Setup

In this project, the general TVB setup was customized to generate two distinct synthetic datasets using Python.

7.1 Region vs. Surface Simulations

Simulations can be executed at two fundamental spatial resolutions. Region-based simulations use a coarse granularity where each parcellated region is simulated as a single homogeneous node. On the other hand, *surface-based simulations* model the cortical surface as a high-resolution 3D mesh. Neural mass models are placed directly on these vertices, utilizing a local connectivity kernel to simulate short-range intra-cortical communication. While computationally heavy, the surface-based approach was strictly adopted in this work because it is necessary to track detailed spatial wave propagation across the cortex and to accurately evaluate the BEM forward model.

7.2 Numerical Integration and Monitors

To compute the temporal evolution of the neural states, a stochastic integration scheme was utilized. Neural systems are intrinsically noisy; thus, a Stochastic Heun integrator was employed. Additive Gaussian white noise was injected into the state variables at each integration step, ensuring a realistic variability in the simulated background activity. The state variables were then monitored and recorded over time to generate the synthetic true source current density $\mathbf{J}_{\text{true}}$.

7.3 Scenario A: The Generic 2D Oscillator

To establish a baseline for spatial accuracy a *Generic 2D Oscillator* was employed to simulate a spatially and time confined cortical activation (a focal Gaussian patch).

This model is a generalization of classic planar dynamical systems and is characterized by a minimal two-dimensional state space:

$$\dot{V} = d\,\tau(-fV^3 + eV^2 + gV + \alpha W + \gamma I) \tag{30}$$

$$\dot{W} = \frac{d}{\tau}\,(cV^2 + bV - \beta W + a) \tag{31}$$

where V represents the aggregated membrane potential (the primary contributor to the EEG signal), W is a slower recovery variable, and I represents the network coupling. By tuning all the parameters, the system can transition between a stable fixed point and limit cycle oscillations. In this specific simulation, the network was coupled using a Linear function, and a focal cortical patch was stimulated to produce a distinct high-amplitude signal against a noisy background.
The resulting spatio-temporal dynamic is illustrated in Figure 1, showcasing the transient signal generated by the target region.

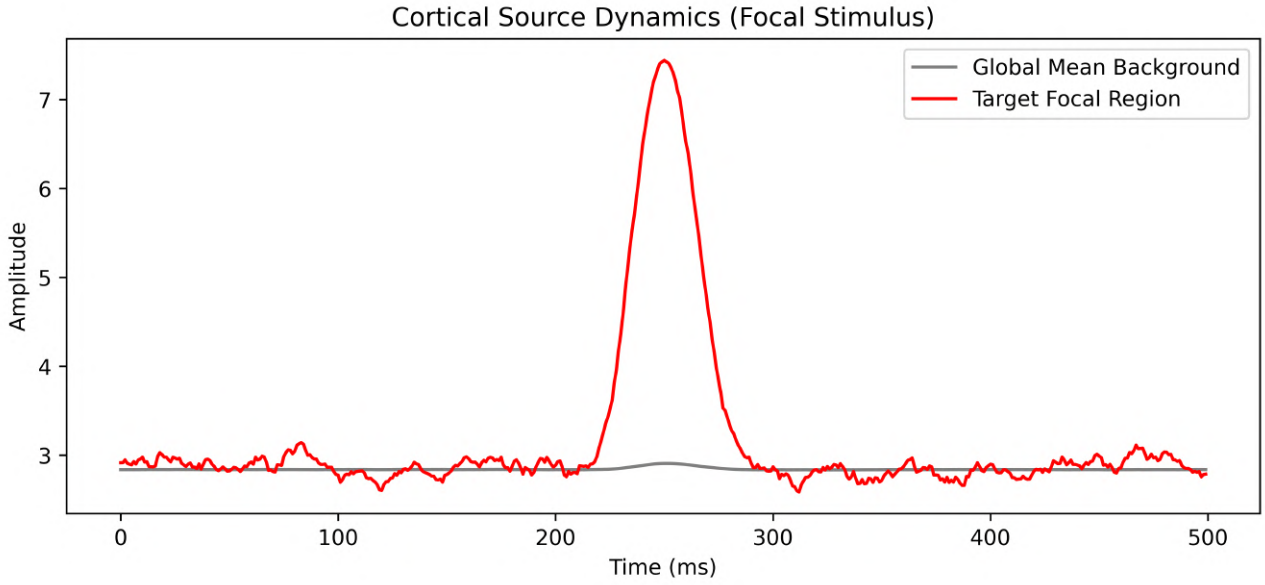


Figure 1: Simulated time series of the Generic 2D Oscillator. The plot highlights the transient activation of the focal target region.

7.4 Scenario B: The Epileptor Model

To test the spatial inversion algorithms against a highly dynamic and clinically relevant scenario, the *Epileptor* phenomenological model was utilized [9]. This simulation focused on the forward generation of a seizure propagation across the cortical surface. The model consists of a six-dimensional state vector governed by the following coupled differential equations:

$$\dot{x}_1 = y_1 - f_1(x_1, x_2) - z + I_1 \tag{32}$$

$$\dot{y}_1 = 1 - 5x_1^2 - y_1 \tag{33}$$

$$\dot{z} = \frac{1}{\tau_0}\,(4(x_1 - x_0) - z) \tag{34}$$

$$\dot{x}_2 = -y_2 + x_2 - x_2^3 + I_2 + 0.002g(x_1) - 0.3(z - 3.5) \tag{35}$$

$$\dot{y}_2 = \frac{1}{\tau_2}\,(-y_2 + f_2(x_2)) \tag{36}$$

The variables x_1, y_1 are responsible for the fast oscillatory discharges typical of the seizure onset, while x_2, y_2 operate on a slower timescale for the spike-and-wave discharges [9]. The two sub-systems are linked by the ultra-slow permittivity variable z, which accumulates during the seizure until it forces the system back to the resting state [8].
Additive noise was injected exclusively into the fast and slow excitability variables, leaving the ultra-slow permittivity variable deterministic.

The complete simulated temporal dynamic, reflecting the transition from the interictal state to the seizure onset and its subsequent termination, is shown in Figure 2.

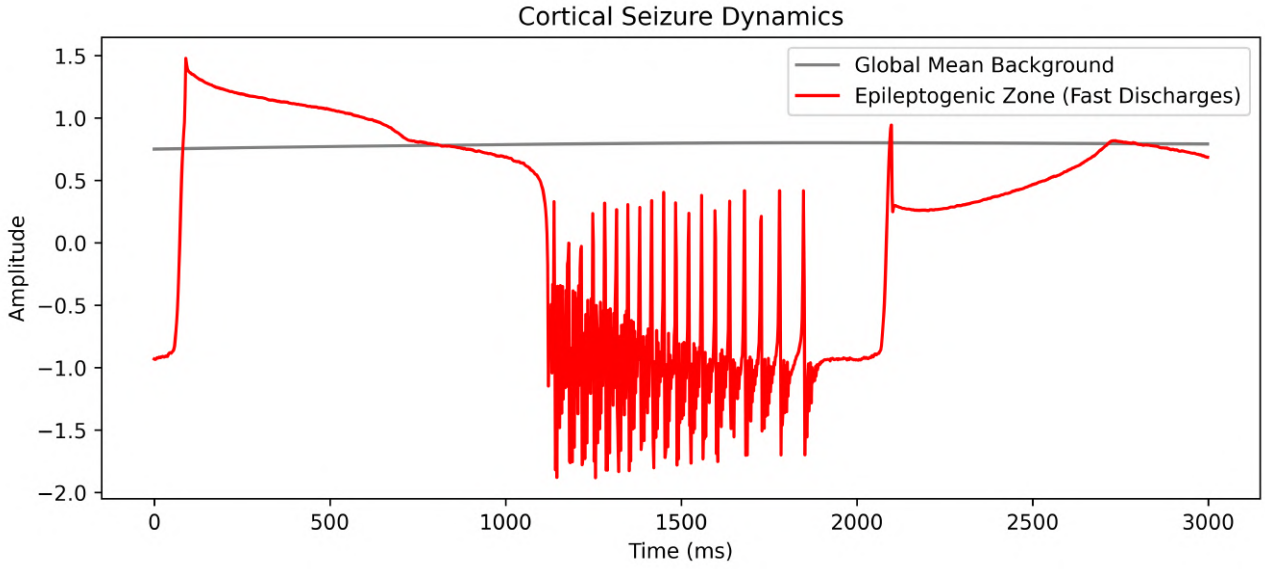


Figure 2: Simulated time series of the Epileptor model. The dynamic transition from the interictal resting state to the onset of fast discharges is clearly visible, followed by slower spike-and-wave patterns, and eventually the seizure termination forced by the ultra-slow permittivity variable z.

8 Validation Framework and Experimental Results

The final objective of this project is to quantitatively validate the accuracy and robustness of the developed Symmetric BEM forward model (the Leadfield matrix $\mathbf{G}$ defined in Section 4). By projecting the synthetic true cortical source activity $\mathbf{J}_{\text{true}}$ to the sensor space, corrupting it with noise, and reconstructing it using spatial inversion algorithms, we can rigorously benchmark how well our matrix $\mathbf{G}$ models the physical volume conduction. In this framework, the inverse problem acts as the ultimate diagnostic tool for the forward formulation.

8.1 Forward Projection and Noise Model

To simulate the macroscopic potentials recorded on the scalp, the true source density generated by TVB is projected through our symmetric Leadfield matrix $\mathbf{G}$. To replicate real clinical recording environments, additive Gaussian white noise $\mathbf{n}$ is injected into the simulated sensor measurements:

$$\mathbf{V}_{\text{sim}} = \mathbf{G}\mathbf{J}_{\text{true}} + \mathbf{n} \tag{37}$$

The severity of the noise is parameterized by the Signal-to-Noise Ratio (SNR). During the evaluation, Monte Carlo simulations were performed in MATLAB. For each SNR level (ranging from highly degraded 0 dB conditions up to clean 25 dB environments), multiple iterations with randomized noise matrices were executed to compute statistically significant error distributions.

8.2 Inverse Algorithms as Validation Tools

To achieve a unique solution and evaluate our matrix $\mathbf{G}$, four linear distributed inverse solutions were compared:

- **Pseudoinverse (PINV):** A naïve baseline method that computes the Moore-Penrose pseudoinverse of the Leadfield matrix, lacking advanced spatial regularization.
- **Minimum Norm Estimate (MNE):** Minimizes the overall energy of the sources using Tikhonov regularization:

$$\hat{\mathbf{J}}_{\text{MNE}} = \mathbf{G}^T(\mathbf{G}\mathbf{G}^T + \lambda\mathbf{I})^{-1}\mathbf{V}_{\text{sim}} \tag{38}$$

- **3. Weighted Minimum Norm Estimate (WMNE):** Counteracts the mathematical depth-bias of standard MNE by introducing a diagonal weighting matrix $\mathbf{W}$ calculated from the column norms of $\mathbf{G}$:

$$\hat{\mathbf{J}}_{\text{WMNE}} = \mathbf{W}^{-1}\mathbf{G}^T(\mathbf{G}\mathbf{W}^{-1}\mathbf{G}^T + \lambda\mathbf{I})^{-1}\mathbf{V}_{\text{sim}} \tag{39}$$

- **4. standardized Low-Resolution Brain Electromagnetic Tomography (sLORETA):** Standardizes the MNE estimate by its theoretical variance $\mathbf{S}_{\hat{J}}$:

$$\mathbf{S}_{\hat{J}} = \mathbf{G}^T(\mathbf{G}\mathbf{G}^T + \lambda\mathbf{I})^{-1}\mathbf{G} \quad \Longrightarrow \quad \hat{\mathbf{J}}_{\text{sLORETA},i} = \mathbf{S}_{\hat{J},ii}^{-1/2}\hat{\mathbf{J}}_{\text{MNE},i} \tag{40}$$

Because the EEG inverse problem is severely ill-posed, the choice of the regularization parameter λ is critical. During the pipeline development, standard heuristic techniques such as the L-curve method and Generalized Cross-Validation (GCV) were initially evaluated. However the optimal λ that showed better results, based on the various simulations, was defined as a function of the Signal-to-Noise Ratio (SNR) and the trace of the Leadfield kernel:

$$\lambda = \frac{\text{tr}(\mathbf{G}\mathbf{G}^T)}{N_c \cdot \text{SNR}_{\text{lin}}} \tag{41}$$

where N_c is the number of EEG channels. This formulation dynamically scales the regularization penalty based on the actual noise level and the specific matrix conditioning, providing the most robust reconstructions across all simulated scenarios.
Because sLORETA accounts for the variance, it theoretically achieves zero localization error under ideal conditions. Therefore, analyzing sLORETA's performance is the most rigorous test to highlight any spatial mapping inaccuracies in our $\mathbf{G}$ matrix.

8.3 Performance Metrics

The reconstructed source maps were evaluated using spatial and temporal metrics:

- **Localization Error (ED1):** The Euclidean distance (in millimeters) between the spatial peak of the ground truth and the estimated peak.

$$ED1 = \left\|\mathbf{r}_{\text{true, peak}} - \mathbf{r}_{\text{est, peak}}\right\|_2 \tag{42}$$

- **Spatial Dispersion (ED2):** Evaluates the "blurriness" of the reconstruction, measuring how much the estimated energy spreads out.

- **Pearson Correlation (Temporal and Spatial):** Evaluates the linear correlation between the temporal envelopes (Hilbert transform) of the true and estimated signals, as well as the spatial map correlations at the signal peaks.

8.4 General Performance of Inverse Algorithms

Before processing the TVB datasets, a standalone validation of the inverse algorithms was performed to establish their baseline behavior utilizing the columns of TVB provided projection matrix.
Figure 3 illustrates the median ED1 and ED2 across different SNR levels. As expected, the unregularized PINV exhibits catastrophic localization errors. Standard MNE and WMNE show improved but consistently high ED1 values due to volume conduction blurring. Conversely, sLORETA demonstrates exceptional accuracy, achieving a median ED1 of 0 mm at 25 dB, proving that the Symmetric BEM matrix $\mathbf{G}$ was assembled correctly. The error only rises gradually as the noise heavily corrupts the signal towards 0 dB.

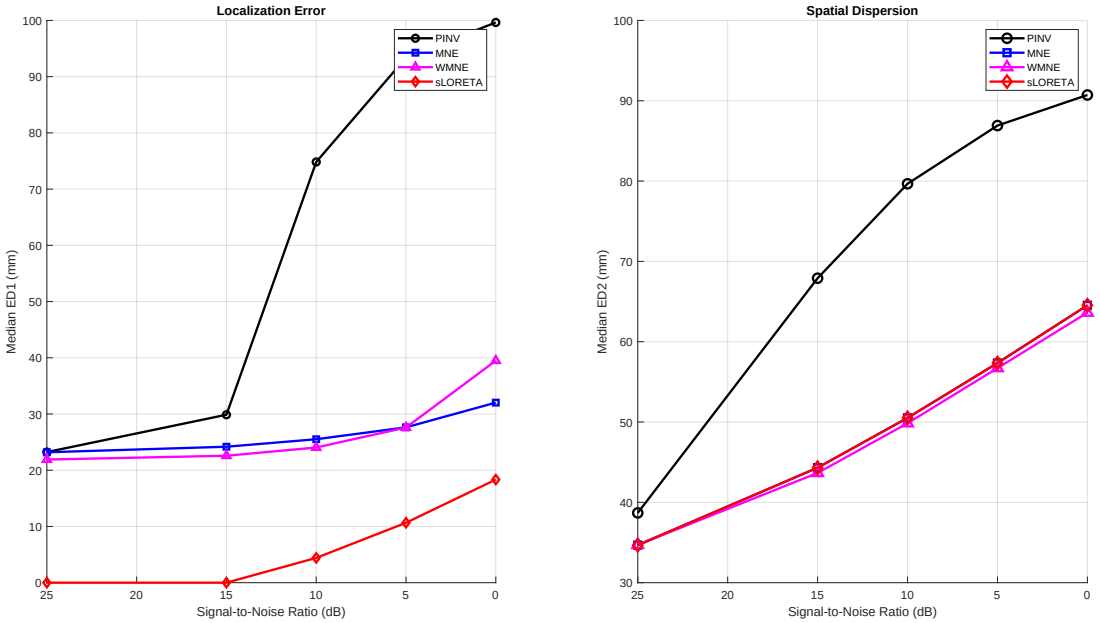


Figure 3: Algorithm comparison. (Left) Localization Error (ED1): sLORETA vastly outperforms other methods. (Right) Spatial Dispersion (ED2): All linear methods exhibit intrinsic blurring that worsens with lower SNR.

To visualize this accuracy, Figure 4 maps the ED1 distribution across the entire cortex for sLORETA at 25 dB. The vast majority of the cortical surface exhibits zero or near-zero errors, confirming the structural integrity of the high-resolution forward model.

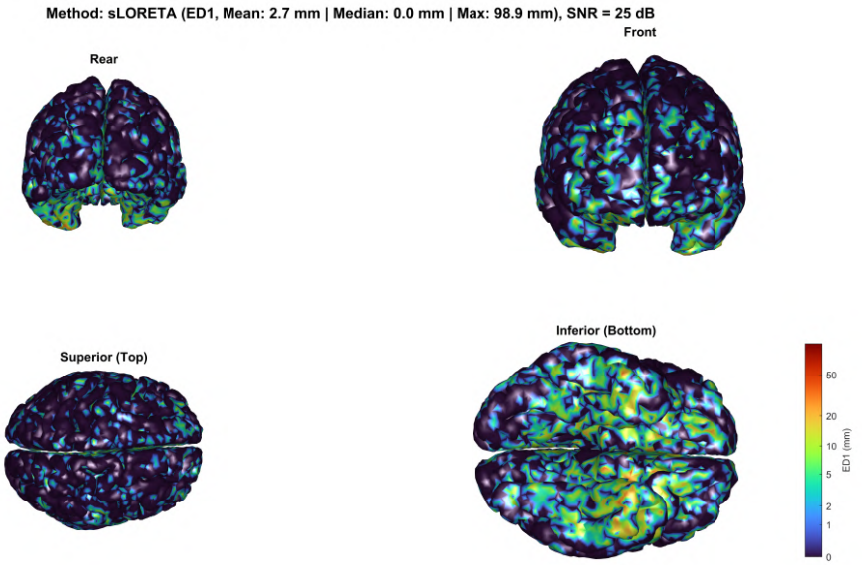


Figure 4: Cortical distribution of the Localization Error (ED1) using sLORETA at 25 dB SNR. The median error is 0.0 mm, demonstrating nearly perfect spatial mapping under low-noise conditions.

To highlight the necessity of using a standardized approach like sLORETA, Figure 5 displays the exact same evaluation performed with the standard MNE algorithm. Even under ideal low-noise conditions (25 dB), MNE suffers from severe spatial inaccuracies. Because it lacks variance standardization, it systematically mislocalizes deeper sources by projecting their energy onto the closest superficial ones.

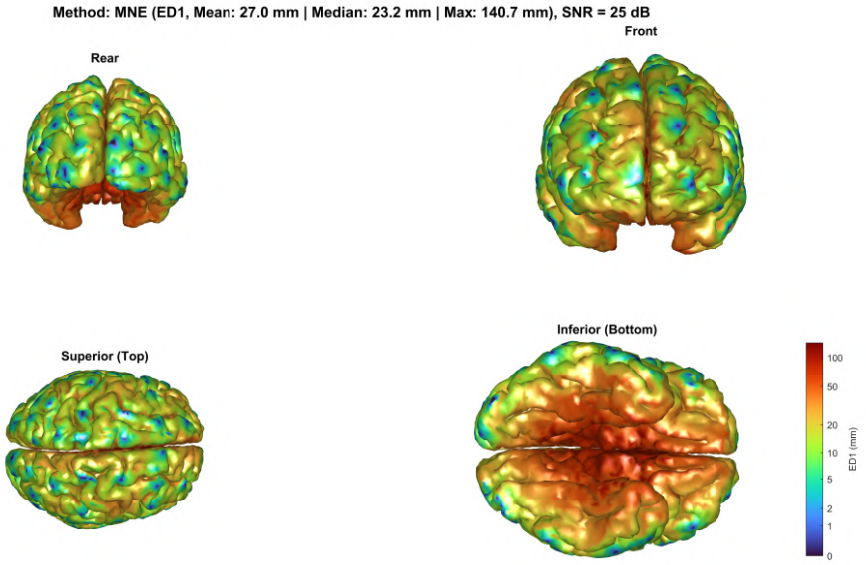


Figure 5: Cortical distribution of the Localization Error (ED1) using standard MNE at 25 dB SNR. The inherent depth bias leads to significant spatial mislocalizations despite the clean signal, visually confirming its inferiority compared to sLORETA.

Finally, to understand the limitations of the pipeline under heavy environmental interference, the sLORETA mapping was evaluated at 0 dB SNR (Figure 6). At this extreme noise level—where the noise power equals the signal power—the algorithm's performance inevitably degrades. However, the degradation is relatively graceful: while the maximum and mean errors increase, the method still manages to reconstruct several source locations with reasonable accuracy, demonstrating the robustness of the combined Symmetric BEM and sLORETA framework.

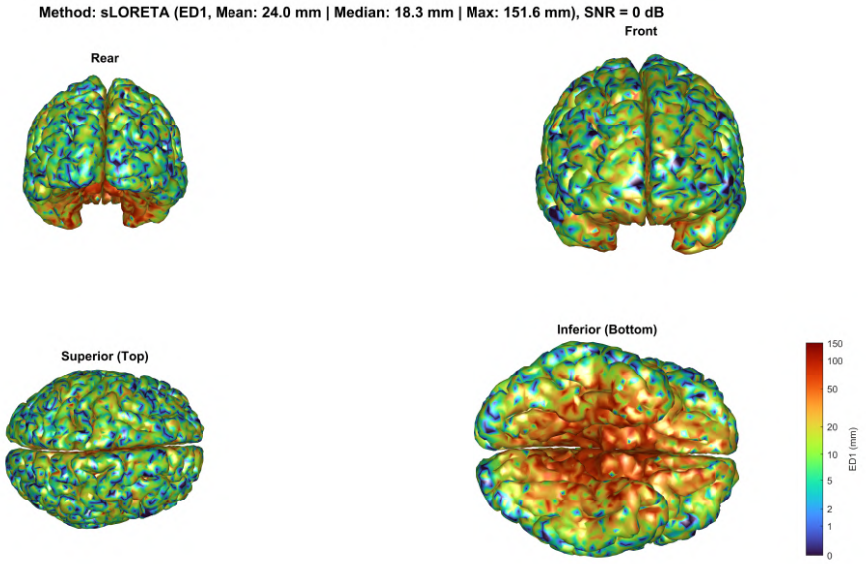


Figure 6: Cortical distribution of the Localization Error (ED1) using sLORETA at 0 dB SNR.

8.5 TVB Validation: Scenario A (Focal Gaussian Activation)

Having established sLORETA as the optimal estimator, the pipeline was applied to the synthetic TVB datasets. Figure 7 displays the successful spatial reconstruction of the TVB Gaussian patch at an SNR of 10 dB.

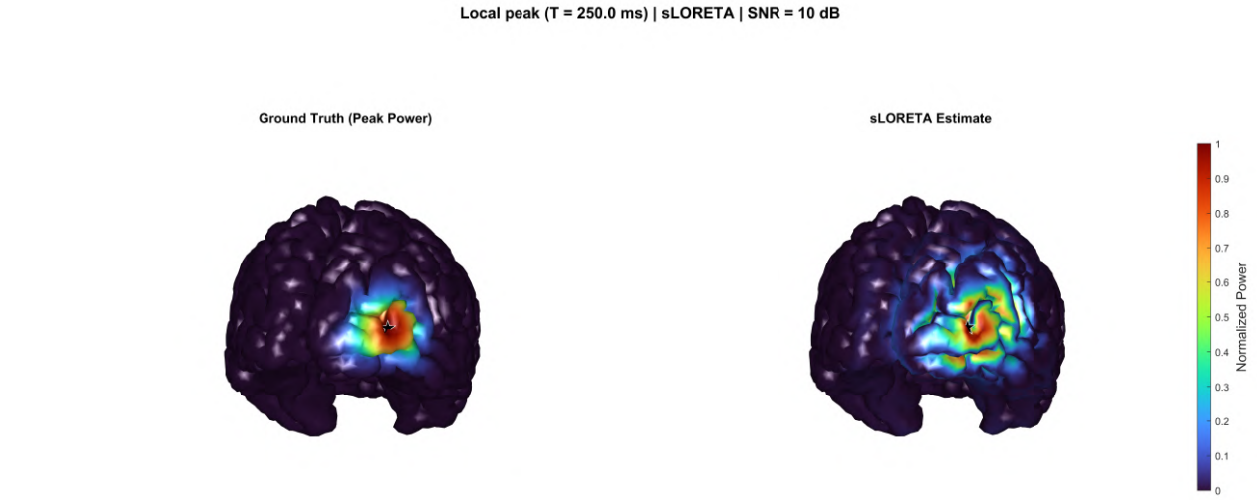


Figure 7: Spatial reconstruction of the TVB Gaussian activation at 10 dB SNR. sLORETA correctly identifies the designated focal target region.

The comprehensive evaluation across all SNR levels for this scenario is shown in Figure 8. Because the Gaussian activation is highly focal and stable, the temporal correlation (Pearson r of the Hilbert Envelope) remains nearly perfect (≈ 1.0) across all noise levels. The energy-weighted ED1 remains extremely low (under 10 mm), confirming that the alignment between the TVB mesh and the BEM mesh was highly precise.

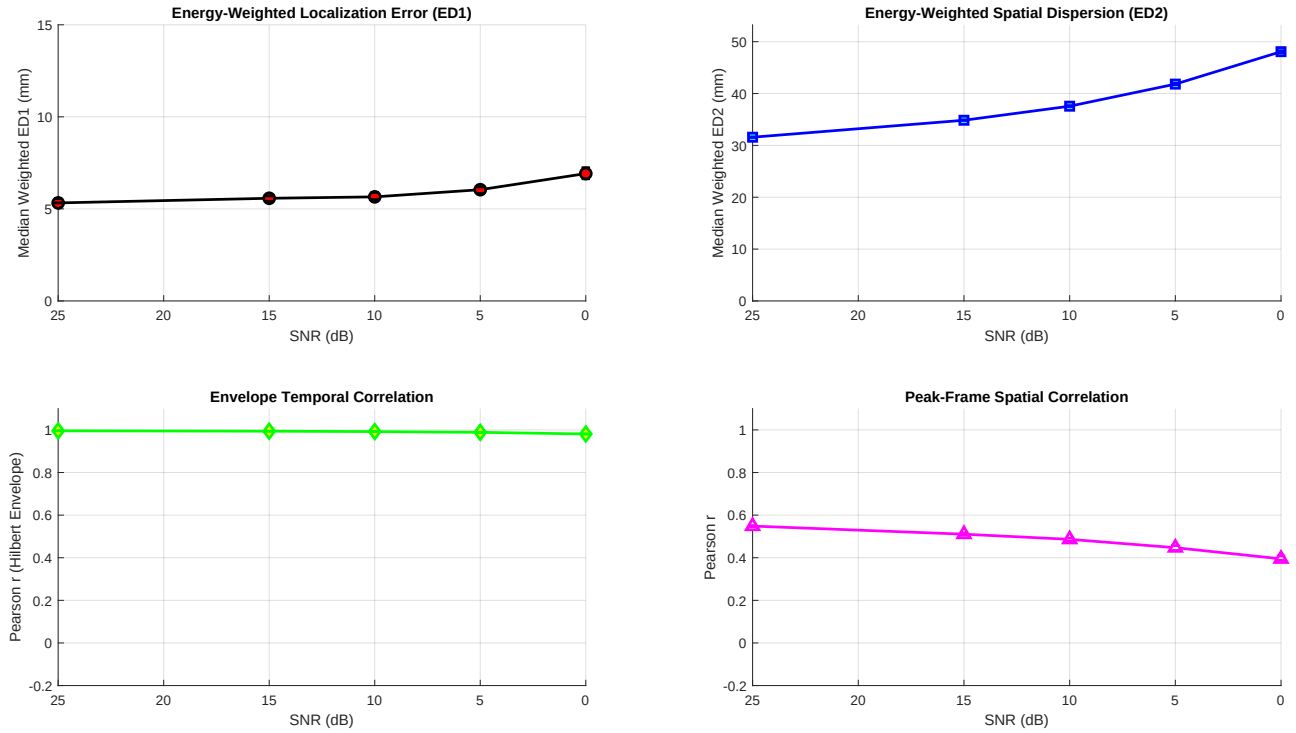


Figure 8: Performance metrics for the TVB Gaussian Activation across SNR levels. The spatial and temporal fidelity of the reconstructed signal remains remarkably stable despite noise injection.

8.6 TVB Validation: Scenario B (Dynamic Seizure Propagation)

The final test challenged the pipeline with the complex spatio-temporal propagation generated by the TVB Epileptor model. Unlike a static patch, a seizure recruits wide cortical areas, generating highly synchronized, distributed signals that heavily stress linear inverse solutions.
Figure 9 shows the reconstructed spatial map of the seizure peak at 10 dB SNR, demonstrating the algorithm's capability to localize the core of the epileptogenic activity despite the broad spatial recruitment.

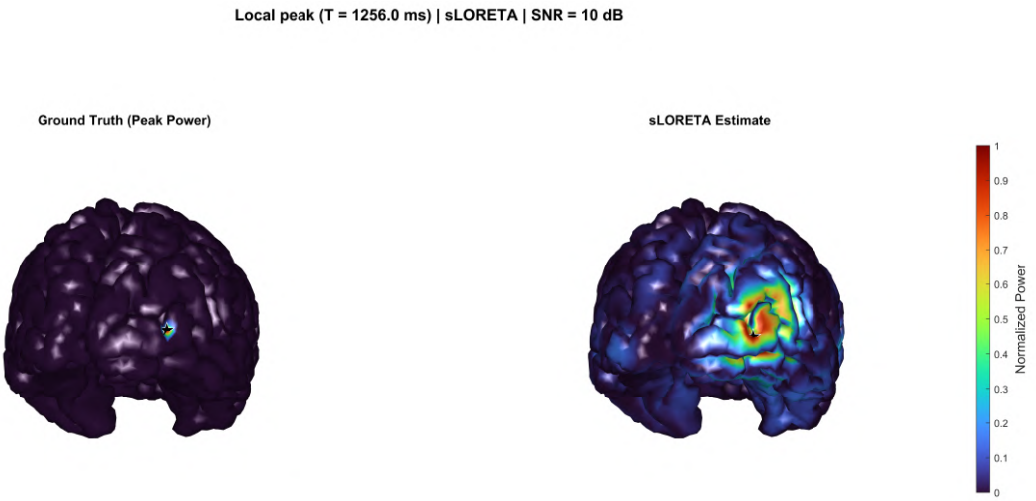


Figure 9: Spatial reconstruction of the TVB Epileptor simulation at 10 dB SNR. The estimated map successfully localizes the peak power of the seizure discharges.

However, the metric analysis in Figure 10 reveals the intrinsic difficulties of this clinical scenario. The ED1 is noticeably higher compared to the Gaussian patch, reflecting the widely distributed nature of the seizure which flattens the energy peaks. Most importantly, while the temporal correlation is decent at high SNRs, it drops drastically at 0 dB. This indicates that heavy background noise critically disrupts the reconstruction of complex, multi-frequency temporal dynamics (like the fast discharges and slow spikes of an epileptic seizure).

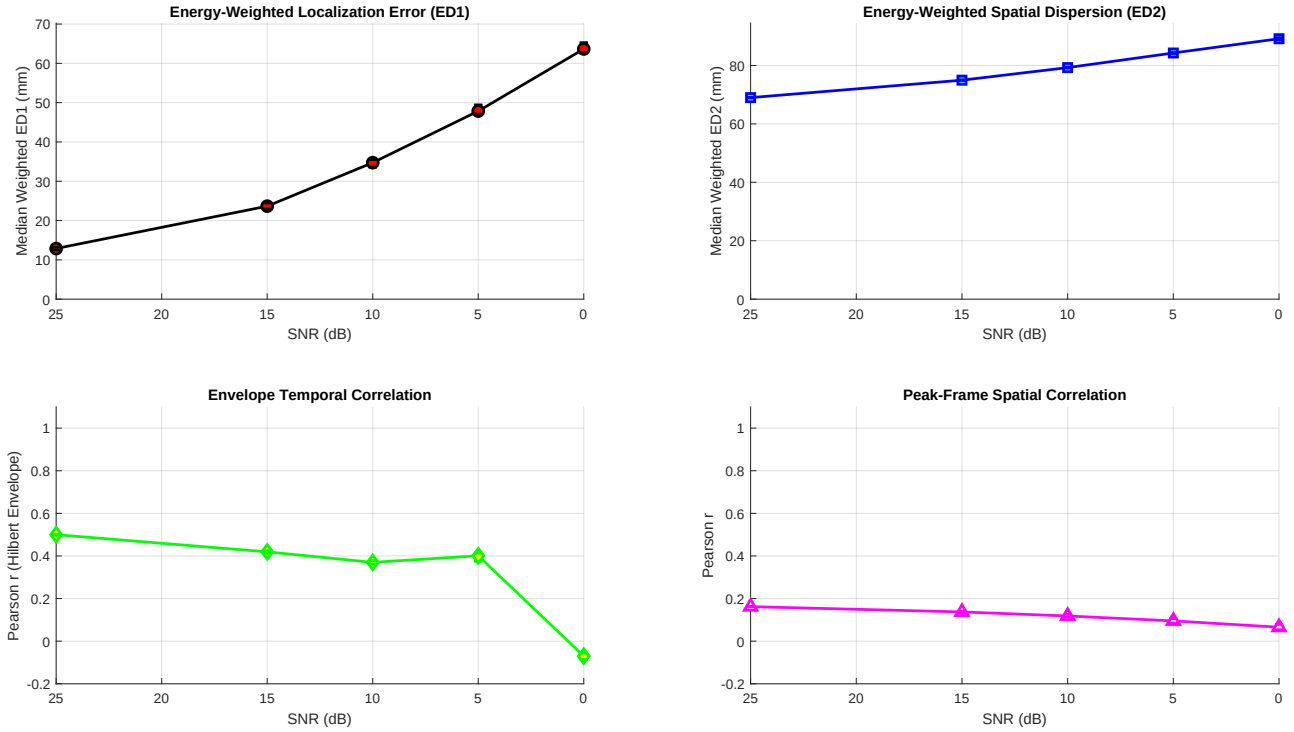


Figure 10: Performance metrics for the TVB Epileptor simulation. The increased spatial dispersion (ED2) and the drop in temporal correlation at low SNR highlight the severe challenges of reconstructing dynamically spreading neural networks.

8.7 Conclusion and Acknowledgments

The comprehensive results presented in this section validate the computational pipeline developed during this internship. The advanced Symmetric BEM formulation provided a highly accurate Leadfield matrix $\mathbf{G}$, demonstrated by sLORETA's zero-error baseline. The subsequent integration with The Virtual Brain proved to be an exceptionally powerful testbed: it highlighted both the robust spatial mapping capabilities of our forward model and the intrinsic temporal limitations of standard inverse algorithms when facing complex, clinically realistic pathologies like epilepsy.

Ultimately, this internship has represented an incredibly rewarding and deeply formative educational experience. I would like to express my sincere gratitude to Professor *Andriulli* for providing me with this opportunity. Furthermore, I am immensely grateful to *Angelo Faccia* from the research team, who closely supervised my work and provided invaluable assistance, constant support, and technical guidance throughout every stage of this project.

References

[1] Hans Hallez et al. "Review on solving the forward problem in EEG source analysis". In: *Journal of NeuroEngineering and Rehabilitation* 4.1 (Nov. 2007). ISSN: 1743-0003. DOI: 10.1186/1743-0003-4-46.

[2] Robert Plonsey and Dennis B. Heppner. "Considerations of quasi-stationarity in electrophysiological systems". In: *The Bulletin of Mathematical Biophysics* 29.4 (Dec. 1967), pp. 657–664. ISSN: 1522-9602. DOI: 10.1007/bf02476917.

[3] David B. Geselowitz. "On Bioelectric Potentials in an Inhomogeneous Volume Conductor". In: *Biophysical Journal* 7.1 (Jan. 1967), pp. 1–11. ISSN: 0006-3495. DOI: 10.1016/s0006-3495(67)86571-8.

[4] J Sarvas. "Basic mathematical and electromagnetic concepts of the biomagnetic inverse problem". In: *Physics in Medicine and Biology* 32.1 (Jan. 1987), pp. 11–22. ISSN: 1361-6560. DOI: 10.1088/0031-9155/32/1/004.

[5] A. Van Oosterom and J. Strackee. "The Solid Angle of a Plane Triangle". In: *IEEE Transactions on Biomedical Engineering* BME-30.2 (Feb. 1983), pp. 125–126. ISSN: 0018-9294. DOI: 10.1109/tbme.1983.325207.

[6] J. Kybic et al. "A common formalism for the Integral formulations of the forward EEG problem". In: *IEEE Transactions on Medical Imaging* 24.1 (Jan. 2005), pp. 12–28. ISSN: 0278-0062. DOI: 10.1109/tmi.2004.837363.

[7] Michael Schirner et al. "An automated pipeline for constructing personalized virtual brains from multimodal neuroimaging data". In: *NeuroImage* 117 (Aug. 2015), pp. 343–357. ISSN: 1053-8119. DOI: 10.1016/j.neuroimage.2015.03.055.

[8] Timothée Proix et al. "Individual brain structure and modelling predict seizure propagation". In: *Brain* 140.3 (Feb. 2017), pp. 641–654. ISSN: 1460-2156. DOI: 10.1093/brain/awx004.

[9] V.K. Jirsa et al. "The Virtual Epileptic Patient: Individualized whole-brain models of epilepsy spread". In: *NeuroImage* 145 (Jan. 2017), pp. 377–388. ISSN: 1053-8119. DOI: 10.1016/j.neuroimage.2016.04.049.

[10] J. Zimmermann et al. "Differentiation of Alzheimer's disease based on local and global parameters in personalized Virtual Brain models". In: *NeuroImage: Clinical* 19 (2018), pp. 240–251. ISSN: 2213-1582. DOI: 10.1016/j.nicl.2018.04.017.

[11] Leon Stefanovski et al. "Linking Molecular Pathways and Large-Scale Computational Modeling to Assess Candidate Disease Mechanisms and Pharmacodynamics in Alzheimer's Disease". In: *Frontiers in Computational Neuroscience* 13 (Aug. 2019). ISSN: 1662-5188. DOI: 10.3389/fncom.2019.00054.

[12] Maria Inez Falcon et al. "Functional Mechanisms of Recovery after Chronic Stroke: Modeling with the Virtual Brain". In: *eneuro* 3.2 (Mar. 2016), ENEURO.0158–15.2016. ISSN: 2373-2822. DOI: 10.1523/eneuro.0158-15.2016.

[13] Hannelore Aerts et al. "Modeling Brain Dynamics in Brain Tumor Patients Using the Virtual Brain". In: *eneuro* 5.3 (May 2018), ENEURO.0083–18.2018. ISSN: 2373-2822. DOI: 10.1523/eneuro.0083-18.2018.